

# Intro to NLP for corpus linguistics

Hakyung Sung  
University of Hawaii at Manoa

August 26, 2026

# Topics to cover

- Course logistics

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- Course logistics
- Intro to course

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- Intro to NLP

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- Intro to NLP and language learning

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- **Instructor:** Hakyung Sung (Prefer to be called Hakyung; she/her)



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  - PhD in Linguistics, MS in Computer Science @ University of Oregon
  - Study how second language learners use words and grammatical structures, using natural language processing approaches
- **Time:** W 3:00-5:50 PM
- **Office hours:**
  - M 1:00-2:00 PM (in-person, Moore 554)
  - By appointment (Email me)
- **Course website:** [https://hksung.github.io/Fall26\\_SLS775/](https://hksung.github.io/Fall26_SLS775/)
  - Link on the Lamakū
- **Email:** [sungh@hawaii.edu](mailto:sungh@hawaii.edu)

## Grading

- Syllabus includes all the details/updates (which is on the course website).

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- (1) Presentation 10%

[https://hksung.github.io/Fall26\\_SLS775/assignments/1\\_paper%20presentation](https://hksung.github.io/Fall26_SLS775/assignments/1_paper%20presentation)

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## Tentative course schedule

| Week | Date | Topic                                                                                      |
|------|------|--------------------------------------------------------------------------------------------|
| 1    | 8/26 | Intro to NLP for corpus linguistics                                                        |
|      |      | <b>Lab:</b> Intro to Python 1                                                              |
| 2    | 9/2  | NLP in learner corpus research<br><a href="#">Presentation 1 (Kyle &amp; Eguchi, 2024)</a> |
|      |      | <b>Lab:</b> Intro to Python 2                                                              |
| 3    | 9/9  | Corpus design and selection<br><a href="#">Presentation 2 (Hanks et al., 2024)</a>         |
|      |      | <b>Lab:</b> A survey of corpora                                                            |

# Grading

- (2) Assignments 4 x 10%

| Week | Date | Topic                                                    | Readings                       | Assignment<br>Due Friday,<br>11:59 pm |
|------|------|----------------------------------------------------------|--------------------------------|---------------------------------------|
| 1    | 8/26 | Intro to NLP for corpus linguistics                      | Dunn (2022)                    | Survey                                |
|      |      | <b>Lab:</b> Intro to Python 1                            | Gries (2026)<br>Meurers (2020) |                                       |
| 2    | 9/2  | NLP in learner corpus research                           | Granger (2026)                 |                                       |
|      |      | <a href="#">Presentation 1 (Kyle &amp; Eguchi, 2024)</a> | Kyle (2021)                    |                                       |
|      |      | <b>Lab:</b> Intro to Python 2                            | Kyle & Sung (2026)             |                                       |
| 3    | 9/9  | Corpus design and selection                              | Gries (2009)                   | Assignment 1                          |
|      |      | <a href="#">Presentation 2 (Hanks et al., 2024)</a>      | Egbert et al. (2022), Ch. 3    |                                       |
|      |      | <b>Lab:</b> A survey of corpora                          |                                |                                       |

# Grading

- (3) **Project 40%**
  - Proposal 3%
  - Midway presentation 5%
  - Midway report 7%
  - Final presentation 10%
  - Final paper 15%



# Grading

- (4) Participation 10%
  - Completing ad-hoc events that the instructor announces in class 2%
  - Active participation in Presentation 4%
  - Active participation in midway/final presentation 4%

# Grading

## Extra credit 2%

You may participate in the Linguistics Beyond the Classroom (LBC) program for an extra 2% credit towards your final course grade. The purpose of this program is for students to get first-hand experience of language research by participating in an on-going research project conducted here at UH. For more information, see <http://ling.lil.hawaii.edu/sites/lbc/>. You may complete either Option 1 (research participation) or Option 2 (500-word event reflection) as outlined on the LBC website. I must receive your completed questionnaire (option 1) or event response (option 2) by December 4<sup>th</sup>, 2026 for you to receive this credit.

## Other policies

**All the information is available in the syllabus:**

- Grading policy (including CR/NC; \*Audit)
- Attendance policy
- Late policy
- Use of AIs
- UH policies and resources

# Expectation

- Ideally, read the assigned readings before/after class.

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# Expectation



# Intro to course

# Natural Language Processing in Corpus Research

# Natural Language Processing in **Corpus Research**

# What counts as corpus research?

Givón (2009)

## Task: Analyzing their discourse contexts

*“What I have done with the [...] CHILDES transcripts [...] is isolate and extract all the MIUs [...] that surround each complex VP construction.”*

### (3) Simple modal interactions

#### a. Deontic: (Eve-I, p. 2)

|                                           |           |
|-------------------------------------------|-----------|
| EVE: Napkin.                              | (request) |
| MOT: Oh, do you <b>want</b> a napkin too? | (offer)   |

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#### c. Deontic: (Eve-I, p. 15-16)

|                                                                    |                              |
|--------------------------------------------------------------------|------------------------------|
| EVE: Bottle.                                                       | (request)                    |
| MOT: What?                                                         | (request for interpretation) |
| EVE: Eve...                                                        | (request)                    |
| MOT: Do you <b>want</b> to taste it?                               | (offer)                      |
| <b>Let's</b> see if Sarah <b>would like</b> to <b>have</b> a drink | (manipulation)               |
| EVE: Eve <b>want</b> some too.                                     | (request)                    |

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2. ~~Anatomy. a body, mass, or part having a special character or function~~
3. a large or complete collection of writings
4. *Linguistics.* a body of utterances, as words or sentences, assumed to be representative of and used for lexical, grammatical, or other linguistic analysis

<https://www.dictionary.com/browse/corpus>

## Is this corpus research?

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## So, what is corpus research?

In a broad sense, if you analyze a **body of utterances** for linguistic purposes, you are doing **corpus research**.

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- **Large** collection? How large it should be?
- Researchers assume that the collection is meaningful and appropriate ( $\approx$  *representative*; will cover this in [Week 3](#)) for the linguistic question being investigated.
- NLP is therefore NOT what makes research **corpus** research.

## Natural Language Processing (NLP) in Corpus Research

# Example

## Givón (2009) Task: Analyzing their discourse contexts

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Is this an NLP approach?

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(request)

- Can we replicate the analysis?
- Can we apply the analysis consistently across datasets?
- Can we scale it up?
- Can NLP make these analyses more systematic and reproducible?

## **Corpus Linguistics**

- Begins with a linguistic question



## Corpus Linguistics

- Begins with a linguistic question
- Examines naturally occurring language

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## Corpus Linguistics

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- Interprets linguistic distributions

## +NLP

- Develops computational methods
- Automatically processes language
- Identifies and predicts linguistic properties
- Replicates findings both within/outside the research group (to argue that your findings were not serendipity but part of science)



## Linguistic question

**Linguistic question**



**Corpus data**

**Linguistic question**



**Corpus data**



**Computational analysis**

**Linguistic question**



**Corpus data**



**Computational analysis**



**Evidence about language use**

**Linguistic question**



**Corpus data**



**Computational analysis**



**Evidence about language use**



**Answer question**

# Two broad computational problems (Dunn, 2022)

## Categorization

What **kind** of thing is this?

- Part-of-speech category
- Grammatical construction
- Register / genre
- Error type
- Proficiency level

## Comparison

How **similar or related** are these?

- Words
- Sentences
- Texts
- Speakers
- Registers

## But the distinction is not about algorithms

Categorization  $\neq$  supervised learning

Comparison  $\neq$  unsupervised learning

- **Supervised learning:** learns from examples with predefined target labels.
- **Unsupervised learning:** identifies patterns or groupings without predefined target labels.

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- will cover in [Week 9 & 11](#)



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- **Supervised learning:** learns from examples with predefined target labels.
- **Unsupervised learning:** identifies patterns or groupings without predefined target labels.
- will cover in [Week 9 & 11](#)

The same problem may be approached in multiple computational ways.

► [Example 1](#)

► [Example 2](#)

# Intro to NLP

# What is NLP?

- NLP
- machine learning
- deep learning
- LLM
- AI
- ...



# What is NLP?

Manning, 2022, pp. 127-128

*“When scientists consider **artificial intelligence**, they mostly think of modeling or recreating the capabilities of an **individual human brain**... The power of **language** is fundamental to human societal intelligence, and language will retain an important role in a future world in which human abilities are augmented by artificial intelligence (AI) tools ... **For these reasons, the field of natural language processing (NLP) emerged in tandem with the earliest development in AI.**”*

Manning, C. D. (2022). Human language understanding & reasoning. Daedalus, 151(2), 127-138.

# What is NLP?

*“NLP is a subfield of **computer science** and artificial intelligence (AI) that uses **machine learning** to enable computers to understand and communicate with human language.”*

<https://www.ibm.com/think/topics/natural-language-processing>

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  - can extract patterns, meanings, and structures that are difficult to capture manually → **Major focus**

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  - leverage nlp-techniques to generate and explore linguistic hypotheses → **Minor focus**



# What is NLP?

- Now, people in many fields are so excited about NLP because they:
  - can extract patterns, meanings, and structures that are difficult to capture manually → **Major focus**
  - leverage nlp-techniques to generate and explore linguistic hypotheses → **Minor focus**
  - enables real-world applications (e.g., search, translation, chatbots, assessment) → NOT our focus

# What can NLP do? (Generally)

## Word level

- Tokenization
- Lemmatization

## Sentence level

- Parsing
- Error detection
- Semantic representation
- Translation

## Text level

- Classification
- Summarization

# What can NLP do? (This class)

## Word level

- Tokenization

Week 4

- Lemmatization

Week 4

# What can NLP do? (This class)

## Word level

- Tokenization

Week 4

- Lemmatization

Week 4

- POS tagging

Week 6

# What can NLP do? (This class)

## Word level

- Tokenization

Week 4

- Lemmatization

Week 4

- POS tagging

Week 6

- Word embeddings

Week 5

# What can NLP do? (This class)

## Word level

- Tokenization  
Week 4
- Lemmatization  
Week 4
- POS tagging  
Week 6
- Word embeddings  
Week 5

## Sentence level

- Dependency  
parsing  
Week 7

# What can NLP do? (This class)

## Word level

- Tokenization  
Week 4
- Lemmatization  
Week 4
- POS tagging  
Week 6
- Word embeddings  
Week 5

## Sentence level

- Dependency parsing  
Week 7
- Linguistic annotation  
Week 9

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## Word level

- Tokenization  
Week 4
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## Sentence level

- Dependency parsing  
Week 7
- Linguistic annotation  
Week 9
- Semantic role labeling  
Week 11



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- Tokenization  
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## Sentence level

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Week 9
- Semantic role labeling  
Week 11

## Text / discourse level

- Linguistic complexity  
Week 8

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Week 6
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- Dependency parsing  
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- Linguistic annotation  
Week 9
- Semantic role labeling  
Week 11

## Text / discourse level

- Linguistic complexity  
Week 8
- Sentiment / discourse analysis  
Week 13

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## Sentence level

- Dependency parsing  
Week 7
- Linguistic annotation  
Week 9
- Semantic role labeling  
Week 11

## Text / discourse level

- Linguistic complexity  
Week 8
- Sentiment / discourse analysis  
Week 13
- Multilingual / multimodal corpora  
Week 14

## Why NLP matters for corpus linguistics

NLP transforms unstructured language into representations that can be systematically searched, counted, compared, and modeled.

## Why NLP may **NOT** matter for corpus linguistics

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**Computational output is not automatically linguistically valid.**

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It is easy to build an NLP pipeline and generate results from ~~corpus~~ data. [▶ data vs. corpus](#)

**Again, computational output is not equal to linguistically valid output!**

# Intro to corpus linguistics

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- These measures can operationalize constructs such as entrenchment and form–meaning associations.

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- Results depend on corpus composition, sampling, and dispersion

# Intro to NLP and language learning

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What NLP task or measure could you use for each question?

# NLP and language learning

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## Language for learners

- How can NLP help us analyze or adapt language presented to learners?

We'll explore this in more detail next week ([Week 2](#)).

## For the course project

Begin with your questions about language.

Linguistic question



Corpus



NLP method



Interpretation

Please complete the **first-class survey**.

# Appendix

## Example: Categorization

Suppose we want to classify texts by register.

- *Supervised*: Train a model using texts already labeled as academic writing, news, or conversation.
- *Unsupervised*: Cluster texts according to their linguistic similarities and examine whether the clusters correspond to recognizable registers.



## Example: Comparison

How similar are the meanings of two sentences?

- Represent sentences as vectors and calculate *cosine similarity*.
- Or train a supervised model using sentence pairs labeled for degrees of semantic similarity.

**Research problem and computational method are not the same thing.**

# Data $\neq$ Corpus (Gries, 2026)

## Language data

- Any collection of linguistic observations
- May or may not have been systematically sampled
- May have limited documentation

## Corpus

- Deliberately constructed
- Systematically organized
- Documented
- Designed to support analysis of a particular population or domain